THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{1}{2}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
2	**(b)** $\frac{1}{2}$	Two equally likely outcomes.
3	**(c)** $\frac{1}{2}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
4	**(a)** $\frac{1}{2}$	Count the multiples of the number.
5	**(d)** 0	Impossible events have probability 0; sure events have probability 1.
6	**(b)** 6	Count all possible outcomes.
7	**(c)** $\frac{5}{7}$	$P(\text{not } E) = 1 - P(E)$.
8	**(b)** $\frac{23}{50}$	Relative frequency $= \frac{\text{times it occurred}}{\text{number of trials}}$.
9	**(b)** $\frac{1}{3}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
10	**(c)** $\frac{1}{2}$	Two equally likely outcomes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Favourable outcomes $= 3$, total outcomes $= 6$ (1); $P = \frac{1}{2}$ (1).	1+1
12	Total balls $= 8$ (1); $P = \frac{5}{8}$ (1).	1+1
13	$P(\text{head}) = \frac{33}{60} = \frac{11}{20}$ (1); $P(\text{tail}) = \frac{27}{60} = \frac{9}{20}$ (1).	1+1
14	Favourable outcomes $= 4$, total outcomes $= 6$ (1); $P = \frac{2}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 10$ (1); $P(\text{red}) = \frac{3}{10}$ (1); $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$ (1).	1+1+1
16	$P(4) = \frac{22}{120} = \frac{11}{60}$ (1); $P(\text{not } 4) = 1 - \frac{11}{60}$ (1); $= \frac{49}{60}$ (1).	1+1+1
17	Multiples of 5: 5, 10, 15, 20, ... (6 numbers) (1); $P = \frac{6}{30} = \frac{1}{5}$ (2).	1+2
	OR: Total $= 9$ (1); $P(\text{red}) = \frac{5}{9}$ (1); $P(\text{not red}) = 1 - \frac{5}{9} = \frac{4}{9}$ (1).	1+1+1
18	$P(6) = \frac{8}{60} = \frac{2}{15}$ (1); $P(\text{not } 6) = 1 - \frac{2}{15}$ (1); $= \frac{13}{15}$ (1).	1+1+1
	OR: Multiples of 4: 4, 8, 12, 16, ... (6 numbers) (1); $P = \frac{6}{24} = \frac{1}{4}$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sample space: HH, HT, TH, TT (4 equally likely outcomes) (1); $P(\text{two heads}) = \frac{1}{4}$ (1); $P(\text{at least one head}) = \frac{3}{4}$ (1); $P(\text{no head}) = \frac{1}{4}$ (1); Check: $P(\text{no head}) = 1 - P(\text{at least one head})$ (1).	1+1+1+1+1
	OR: Total outcomes $= 6$ (1); $P(\text{an even number}) = \frac{3}{6} = \frac{1}{2}$ (1); $P(\text{a number greater than } 4) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 4 outcomes (1); $P = \frac{4}{6} = \frac{2}{3}$ (1).	1+1+1+1+1
20	Total balls $= 15$ (1); $P(\text{red}) = \frac{4}{15}$, $P(\text{blue}) = \frac{1}{3}$ (1); $P(\text{green}) = \frac{2}{5}$ (1); Sum $= \frac{4+5+6}{15} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Total outcomes $= 6$ (1); $P(\text{an odd number}) = \frac{3}{6} = \frac{1}{2}$ (1); $P(\text{a number less than } 3) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 4 outcomes (1); $P = \frac{4}{6} = \frac{2}{3}$ (1).	1+1+1+1+1

Mistakes to avoid:

- Giving a probability greater than 1 or a negative value, or leaving it as a count instead of a fraction.
- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(a) $\frac{1}{3}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
2	(b) $\frac{1}{5}$	Count the multiples of the number.
3	(b) 1	Impossible events have probability 0; sure events have probability 1.
4	(c) 2	Count all possible outcomes.
5	(c) $\frac{5}{8}$	$P(\text{not } E) = 1 - P(E)$.
6	(c) $\frac{27}{50}$	Relative frequency $= \frac{\text{times it occurred}}{\text{number of trials}}$.
7	(d) $\frac{1}{3}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
8	(a) $\frac{3}{8}$	Favourable balls over total balls.
9	(d) $\frac{2}{5}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
10	(b) $\frac{1}{3}$	Count the multiples of the number.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total balls $= 15$ (1); $P = \frac{3}{5}$ (1).	1+1
12	$P(\text{head}) = \frac{72}{150} = \frac{12}{25}$ (1); $P(\text{tail}) = \frac{78}{150} = \frac{13}{25}$ (1).	1+1
13	Favourable outcomes $= 2$, total outcomes $= 6$ (1); $P = \frac{1}{3}$ (1).	1+1
14	Total balls $= 15$ (1); $P = \frac{7}{15}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 20$ (1); $P(\text{red}) = \frac{2}{5}$ (1); $P(\text{not red}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1).	1+1+1
16	$P(1) = \frac{17}{90} = \frac{17}{90}$ (1); $P(\text{not } 1) = 1 - \frac{17}{90}$ (1); $= \frac{73}{90}$ (1).	1+1+1
17	Multiples of 8: 8, 16, 24, 32, ... (5 numbers) (1); $P = \frac{5}{40} = \frac{1}{8}$ (2).	1+2
	OR: Total $= 15$ (1); $P(\text{red}) = \frac{3}{5}$ (1); $P(\text{not red}) = 1 - \frac{3}{5} = \frac{2}{5}$ (1).	1+1+1
18	$P(3) = \frac{28}{150} = \frac{14}{75}$ (1); $P(\text{not } 3) = 1 - \frac{14}{75}$ (1); $= \frac{61}{75}$ (1).	1+1+1
	OR: Multiples of 6: 6, 12, 18, 24, ... (6 numbers) (1); $P = \frac{6}{36} = \frac{1}{6}$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total balls $= 15$ (1); $P(\text{red}) = \frac{1}{5}$, $P(\text{blue}) = \frac{7}{15}$ (1); $P(\text{green}) = \frac{1}{3}$ (1); Sum $= \frac{3+7+5}{15} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Total outcomes $= 6$ (1); $P(\text{a multiple of } 3) = \frac{2}{6} = \frac{1}{3}$ (1); $P(\text{a number at most } 2) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 4 outcomes (1); $P = \frac{4}{6} = \frac{2}{3}$ (1).	1+1+1+1+1
20	Total balls $= 18$ (1); $P(\text{red}) = \frac{1}{3}$, $P(\text{blue}) = \frac{4}{9}$ (1); $P(\text{green}) = \frac{2}{9}$ (1); Sum $= \frac{6+8+4}{18} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Total outcomes $= 6$ (1); $P(\text{a prime number}) = \frac{3}{6} = \frac{1}{2}$ (1); $P(\text{a perfect square}) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 5 outcomes (1); $P = \frac{5}{6} = \frac{5}{6}$ (1).	1+1+1+1+1

Mistakes to avoid:

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- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0	Impossible events have probability 0; sure events have probability 1.
2	**(d)** 4	Count all possible outcomes.
3	**(c)** $\frac{4}{5}$	$P(\text{not } E) = 1 - P(E)$.
4	**(b)** $\frac{9}{20}$	Relative frequency $= \frac{\text{times it occurred}}{\text{number of trials}}$.
5	**(c)** $\frac{1}{3}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
6	**(d)** $\frac{7}{11}$	Favourable balls over total balls.
7	**(a)** $\frac{2}{3}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
8	**(a)** $\frac{1}{5}$	Count the multiples of the number.
9	**(a)** 1	Impossible events have probability 0; sure events have probability 1.
10	**(d)** 8	Count all possible outcomes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$P(\text{head}) = \frac{35}{80} = \frac{7}{16}$ (1); $P(\text{tail}) = \frac{45}{80} = \frac{9}{16}$ (1).	1+1
12	Favourable outcomes $= 2$, total outcomes $= 6$ (1); $P = \frac{1}{3}$ (1).	1+1
13	Total balls $= 15$ (1); $P = \frac{11}{15}$ (1).	1+1
14	$P(\text{head}) = \frac{120}{250} = \frac{12}{25}$ (1); $P(\text{tail}) = \frac{130}{250} = \frac{13}{25}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 18$ (1); $P(\text{red}) = \frac{2}{9}$ (1); $P(\text{not red}) = 1 - \frac{2}{9} = \frac{7}{9}$ (1).	1+1+1
16	$P(5) = \frac{12}{80} = \frac{3}{20}$ (1); $P(\text{not } 5) = 1 - \frac{3}{20}$ (1); $= \frac{17}{20}$ (1).	1+1+1
17	Multiples of 10: 10, 20, 30, 40, ... (5 numbers) (1); $P = \frac{5}{50} = \frac{1}{10}$ (2).	1+2
	OR: Total $= 15$ (1); $P(\text{red}) = \frac{2}{3}$ (1); $P(\text{not red}) = 1 - \frac{2}{3} = \frac{1}{3}$ (1).	1+1+1
18	$P(2) = \frac{36}{200} = \frac{9}{50}$ (1); $P(\text{not } 2) = 1 - \frac{9}{50}$ (1); $= \frac{41}{50}$ (1).	1+1+1
	OR: Multiples of 9: 9, 18, 27, 36, ... (5 numbers) (1); $P = \frac{5}{45} = \frac{1}{9}$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total balls $= 15$ (1); $P(\text{red}) = \frac{2}{15}$, $P(\text{blue}) = \frac{3}{5}$ (1); $P(\text{green}) = \frac{4}{15}$ (1); Sum $= \frac{2+9+4}{15} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Total outcomes $= 6$ (1); $P(\text{the number } 5) = \frac{1}{6} = \frac{1}{6}$ (1); $P(\text{a number less than } 3) = \frac{2}{6} = \frac{1}{3}$ (1); Favourable for 'or': 3 outcomes (1); $P = \frac{3}{6} = \frac{1}{2}$ (1).	1+1+1+1+1
20	Total balls $= 20$ (1); $P(\text{red}) = \frac{1}{4}$, $P(\text{blue}) = \frac{3}{10}$ (1); $P(\text{green}) = \frac{9}{20}$ (1); Sum $= \frac{5+6+9}{20} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Every outcome (1 to 6) is less than 7: 6 favourable of 6 (1); $P = \frac{6}{6} = 1$ (sure event) (1); No outcome is 0: 0 favourable (1); $P = \frac{0}{6} = 0$ (impossible event) (1); A probability lies between 0 and 1 (1).	1+1+1+1+1

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